J. of the Acad. Mark. Sci. (2011) 39:467–479
DOI 10.1007/s11747-010-0204-7

BRIEF REPORT

Benefits and challenges of conducting multiple methods research in marketing

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Received: 18 May 2009 / Accepted: 20 May 2010 / Published online: 16 June 2010
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Abstract Using multiple methods to study a phenomenon is proposed to produce results that are more robust and compelling than single method studies. This research note investigates the implementation of multiple methods research in marketing. In the main study, we conducted a content analysis of articles published in five leading marketing journals from 1990 to 2008: *Journal of the Academy of Marketing Science, Journal of Consumer Research, Journal of Marketing, Journal of Marketing Research, and Marketing Science*. To aid in interpretation of findings from the content analysis, we gathered verbal data from authors of recent multiple methods studies to gain further insight into the benefits and challenges of conducting and publishing multiple methods research.

Keywords Research methods · Research design · Multiple methods research

The authors wish to acknowledge the research assistance provided on this project by Sunny Choi and Han Mo Oh who are PhD students at Texas Tech University.

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Marketing research is often criticized for its lack of diversity in research methods and failure to employ additional methods that maybe more appropriate for investigation of dynamic, complex phenomena (e.g., Deshpande 1983; Hirschman 1986; Hudson and Ozanne 1988). Traditionally, marketing research has largely relied on single-method quantitative research design, such as experiments, surveys, or modeling of secondary data. Adhering to a limited set of preferred methods introduces "certain inherent methods biases" in the discipline and delimits the scope of research by confining inquiry to only those aspects of phenomena that are amenable to the methods selected (Deshpande 1983). Marketing phenomena are manifestations of the multifaceted behavior of human beings as individuals and in groups; hence, "no single method of research fully captures any particular phenomenon" (Tellis et al. 1999, p. 121). Different methods produce different insights, but this does not mean that one method is superior to another. Rather it points to the danger of overreliance on a few preferred methods. The use of multiple methods to investigate the same phenomenon can produce robust findings that are "far more compelling than single method outcomes" (Stewart 2009, p. 382). However, little is known about the implementation of multiple methods research in marketing. To what extent is multiple methods design employed in marketing research? What are the benefits of using multiple methods design? What difficulties might researchers encounter in conducting and publishing multiple methods studies?

To answer these research questions, we investigate the implementation of multiple methods research in marketing in a content analysis of five leading marketing journals and a survey of authors of multiple methods studies. The following section defines multiple methods research, discusses the design of multiple methods studies, and provides a taxonomy of four basic multiple methods research designs. Next, we provide

examples of the four designs from the marketing literature and examine the research purpose for each design. We then describe our research method and report the findings. We conclude with a discussion of implications for marketing research.

Multiple methods research

Consistent with prior definitions, we define *multiple methods research* as the type of research in which researchers draw on data from more than one source and employ more than one type of analysis (Creswell and Plano Clark 2007; Johnson and Onwuegbuzie 2004). Thus, multiple methods studies may employ two or more qualitative methods, two or more quantitative methods, or a combination of qualitative and quantitative methods in what is called a mixed methods approach. It is important to note that studies using a single data source and multiple methods (e.g., a survey with scaled and open-ended responses) do not meet the criteria for multiple methods research. Similarly, studies that employ multiple data sources and a single method (e.g., multiple experiments) are not considered multiple methods research.

A series of steps advanced multiple methods research (MMR), beginning with the introduction of the multitrait-multimethod (MTMM) matrix proposed by Campbell and Fiske (1959) as a way to assure that observed variance is actually attributable to the trait under study rather than the method used. The introduction of the MTMM matrix stimulated the use of multiple quantitative techniques in the same study. Subsequently, Denzin (1978) applied the term *triangulation* to describe investigations that combine multiple data sources and multiple methods. Triangulation is further explicated by Jick (1979), who distinguished within-methods triangulation (i.e., multiple quantitative or multiple qualitative methods) from across-methods triangulation (i.e., combining qualitative and quantitative methods). Patton (1990) extended consideration of triangulation to analyses across multiple investigators, leading to the popularization of the use of two or more qualitative techniques in a single study. Recent work advocates combining quantitative and qualitative methods in a *mixed methods* research design (e.g., Creswell 2003; Johnson and Onwuegbuzie 2004; Tashakkori and Teddlie 1998). In a recent reflection on the status of mixed methods research, Tashakkori (2010) uses the terminology of *integrated* research design to emphasize the importance of interweaving various methods at different stages in the research process.

Tashakkori and Teddlie (1998) describe four basic research purposes for mixed methods research that we adopt for the current study: (1) *development,* or use of one study to inform a subsequent study; (2) *initiation,* or the use of a preliminary study to launch the main study; (3) *complementarity*, or concurrent examination of various facets of a phenomenon through two or more studies, and; (4) *interpretation,* or the concurrent use of a second study to explain or confirm the results of the main study. Note that two fundamental design decisions are the *weight* assigned to each method and the *timing* of the use of methods (Creswell and Plano Clark 2007; Johnson and Onwuegbuzie 2004; Tashakkori and Teddlie 1998). We use these two considerations to construct a matrix that displays four basic MMR designs and their related research purposes (see Fig. 1).

The weight dimension describes the relative level of reliance on each method. The researcher may decide to assign either equal or unequal weights to the methods, reflecting their relative importance to answering the research question. The timing dimension reflects the temporal order of the use of the data in answering the research question and is classified as sequential or concurrent (Creswell and Plano Clark 2007). Sequential timing means that the first method is employed to collect, analyze, and interpret the first data set before the second method is implemented. Hence, there are two or more phases in sequential research designs. Concurrent timing means that multiple sets of data are collected, analyzed, and interpreted at approximately the same time, in a one-phase study.

Examples of multiple methods research in marketing

Development

When equally weighted methods are implemented sequentially, the second method is used for the *development* of findings from the first method. The researcher's intent is to use the results obtained from the first method to inform a subsequent study, thereby expanding the insights generated about the research problem. Results are reported separately for each phase of the study, followed by a general discussion that ties them together by comparing and contrasting findings. For example, Venable et al. (2005) examine brand personality in the context of nonprofit brands. They conduct six studies using a development research design. In phase one, they gather qualitative data in three studies (two focus groups and depth interviews) to discover the dimensions of brand personality in this new context. In phase two, the dimensions identified in phase one are used to develop items for use in three surveys. The data collected from the surveys are then used in exploratory and confirmatory factor analyses to develop measures of brand personality specifically for the nonprofit context and to test the influence of brand personality on charitable donations. The qualitative studies provide evidence for previously unidentified dimensions of brand personality that are subsequently operationalized and tested in quantitative studies.

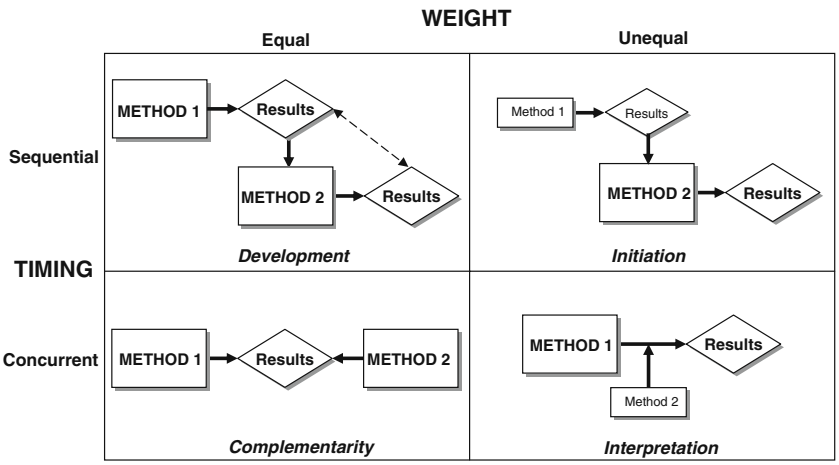


Figure 1 Research purposes in multiple methods research designs. Adapted from Tashakkori and Teddlie (1998).

Initiation

Similar to the development design, the intent of the initiation design is to use results from an initial study to inform a second study that uses a different method. However, the initiation design differs from the development design in that the methods have unequal weights. The less heavily weighted method is employed to initiate the research and is secondary to the primary method that is used in the main study. Results are reported separately, but the focus of the discussion is on the main study. The initiation design is adopted when there is a need for preliminary exploration of the phenomenon because there are no guiding frameworks or theory to identify relevant variables or predict relationships among variables (Creswell and Plano Clark 2007; Tashakkori and Teddlie 1998). Dahl and Moreau (2002) adopt an initiation design to examine cognitive processes in new product ideation. They initiate the research by using verbal protocols to understand of the role of analogy in new product idea generation and to develop hypothesized relationships among variables. Then they conduct experiments to manipulate key factors identified in the initial study. They find that the extent of the analogy transfer, the types of analogies used, and the presence of external primes have a positive influence on new product design.

Complementarity

The purpose of the complementarity design is to examine different, but complementary, aspects of the same phenomenon. In this design, the researcher employs equally weighted methods in a one-phase design; that is, the researcher analyzes and interprets the data from multiple studies concurrently, merging the findings in a single report of results. Unlike the development and initiation designs, the sequencing of methods is not relevant to answering the research questions; that is, results achieved using one method do not inform the design or implementation of subsequent methods. The data maybe collected concurrently (e.g., simultaneous participant observation and interviews) or sequentially (e.g., content analysis followed by surveys). However, the data are analyzed and interpreted in a single report of results. Coulter et al. (2003) combine a survey with depth interviews of women in Budapest and Romania to understand product involvement and brand commitment in the context of postsocialist societies. Results are integrated in a discussion of insights gained through the consumer cases supplemented by survey findings. Their findings link prominent political-cultural discourses, cultural intermediaries, social networks and personal life themes with involvement with branded products.

Interpretation

The interpretation design is used when the researcher anticipates the need to explain or confirm findings from the main study. It is a one-phase, concurrent design in which a secondary method is employed to interpret or support results obtained from the primary method. Similar to the complementarity design, the data maybe collected sequentially, but they are used concurrently in the interpretation and report of findings. Unlike the complementarity design, the methods are not equally weighted. Commuri and Gentry (2005) examine household choice when wives earn more money than their husbands. In the main study, they use an extended case method that involves 20 wife-as-chief-wage-earner (WCWE) couples over a 2-year period. The purpose of the study is to determine patterns and drivers for resource allocation in WCWE households. In the main study, they find that in WCWE households, joint pools of money are used to cover routine expenses, but separate pools are used for various reasons. They then conduct an online survey of 122 married adults to confirm differences revealed in the primary study in

patterns in resource allocation across WCWE and MCWE (man-as-chief-wage-earner) couples.

Study method

To examine the implementation of multiple methods research in marketing, we employed an interpretation design (i.e., unequal weight-concurrent timing). In the main study, we conducted a content analysis of articles published between 1990 and 2008 in five leading marketing journals. To aid the interpretation of findings in the content analysis, we collected verbal data from authors of recently published multiple methods studies to confirm our classification scheme and gain insight into the benefits and challenges of conducting and publishing MMR.

Data collection

In the content analysis, we follow Tellis et al. (1999) in selecting *Journal of Marketing, Journal of Marketing Research, Journal of Consumer Research,* and *Marketing Science* (*JM*, *JMR*, *JCR*, and *MKS*, respectively) as a good representation of marketing journals that are recognized as publishing high quality, relevant marketing research. In addition, we included the *Journal of the Academy of Marketing Science* (*JAMS*) because it is used in other journal reviews (e.g., Svensson 2006), and it is the leading journal of a major international marketing organization, the Academy of Marketing Science. In addition, the impact factor of *JAMS* has been ranked in the top five with the other four journals during the period of our review (Stewart 2007; Zinkhan 2003).

We gathered verbal data in an email survey of authors of recent multiple methods studies identified in the content analysis. Following Dillman (2000), 42 authors were sent emails introducing the study, followed by email delivery of the questionnaire 1 week later. A second questionnaire was sent to non-respondents 2 weeks after the first questionnaire delivery. Open-ended questions were used to verify the research team's classification of the weight and timing of methods used in the studies and to elicit reports of the benefits and challenges of MMR. We contacted authors of more recent studies (i.e., published after 2000) in order to limit concerns about the authors' ability to accurately recall their experiences. Responses were received from 37 authors, providing a response rate of 88%.

Data analysis

A five-member research team reviewed every article published in the five journals from 1990 to 2008. Following Stremersch et al. (2007), each researcher reviewed a subsample of the articles, and all followed the same classification procedure. Team members checked each article starting with the abstract, followed by the introduction and research method section, and including the entire article, if necessary. Articles with less than four pages and those labeled as a reflection or review (e.g., software or book reviews) were excluded, yielding a total sample of 3,289 articles. Articles were first classified by their content as conceptual or empirical. Empirical articles (2,733 in total) were then examined to determine the number of data sources and the specific research method(s) employed. Articles that used multiple methods as defined here (i.e., more than one data source and more than one method) were flagged for further analysis and classification by research design and purpose. This information, along with journal, volume, issue, and authors, was logged into spreadsheets and compiled when the reviews were completed (Appendix 1).

To check classification reliability, a randomly drawn subset of 66 articles (2% of the sample) was verified by a researcher different from the original coder (Krippendorff 1980; Stremersch et al. 2007). Coding was verified on four classification criteria for each article: (1) conceptual or empirical categorization, (2) number of data sources, (3) quantitative methods employed, and (4) qualitative methods employed. Overall, there were five disagreements found in the reliability check. There was disagreement on two articles that used secondary data to support conceptual models, with one coder classifying the articles as conceptual and the other as empirical. There were discrepancies in the number of data sources logged for three articles, which did not affect their classifications as multiple methods studies. There were no disagreements on the classification of methods as quantitative or qualitative. Thus, the reliability check yielded a 98% match with the initial classifications. All studies that were flagged as multiple methods were independently reviewed for classification reliability. Twenty-six articles were initially coded as MMR because they mentioned preliminary studies that used different methods, typically interviews or focus groups, and different data sources from those employed in the main study. However, the studies could not be classified as MMR in the final analysis because they did not fully report the data, analyses, or findings from those preliminary studies. Therefore, the reliability of classification of multiple methods studies was 79%. Both reviews surpassed the level deemed acceptable for similar types of analyses (Stremersch et al. 2007).

Verbal data collected in the survey of authors were saved as individual text documents and loaded into NVivo software for analysis (QSR International 2008). Each document was independently coded by two researchers for themes of meaning following the coding scheme in Appendix 2 (Krippendorff 1980; Miles and Huberman 1994). The unit of analysis was a single assertion about a subject, which ranged from a few words up to an entire paragraph. The

themes were used to interpret authors' perceptions of the benefits and challenges of MMR. Themes were compared across research designs and phases of conducting research to draw further insights. The quality of the analysis was determined by assessing reliability and validity (Miles and Huberman 1994). Interrater reliability was calculated by the two researchers who coded the documents. There was agreement on 346 out of 438 units of analysis coded (79%) across the 37 documents, and discrepancies were resolved by gaining agreement between the coders. The analysis was summarized in a white paper that was reviewed by respondents in a member check of the validity of the interpretation of responses. Validity was further demonstrated through the combination of the convergence of data patterns from multiple sources and the chain of evidence from the study protocol to the results (Hirschman 1986).

Study results

Prevalence of multiple methods research

Multiple methods studies represent 2.9% (95 of 3,289) of the total number of articles and 3.5% (95 of 2,733) of the empirical articles. The percentages of empirical articles that report multiple methods studies are fairly consistent across the five journals (*JAMS* 2.5%; *JMR* 3.2%; *MKS* 3.6%; *JCR* 3.8%; *JM* 4.2%). By comparison, 5% of studies in the top five journals in sociology and 7% of the top five journals in psychology combine quantitative and qualitative methods in mixed methods research designs (Alise and Teddlie 2010), which is a sub-category of MMR as defined in the present research. There appears to be no trend over time, with the percentage of multiple methods studies each year ranging from 2% to 6%. However, there was a spike in 2002 (12.6%) attributable to higher numbers of studies published in *JM* and *JMR*. Fifty-one of the 95 multiple methods studies (54%) incorporate multiple quantitative methods, 41 (43%) are mixed methods studies, and only 3 (3%) use multiple qualitative methods. The 95 multiple methods studies are grouped by research design in Appendix 1.

Research design

The majority (60%) of the multiple methods studies identified in the content analysis use a development research design. According to the authors surveyed, this design was implemented in order to meet the need to triangulate methods to offset inherent weaknesses of single methods: "Lab experiments are great for internal validity, field data are better for external validity." Development design was often used to understand or test a process: "The survey ... was the best way to test our general hypothesis, but not a great way to test underlying processes." Such studies typically used an experimental method for the second phase of research in order to further investigate the phenomenon in a controlled environment. The development design was also used to "rule out alternative explanations" through additional theory testing and to "help think about boundary conditions" for the phenomenon.

Twenty percent of the studies use unequally weighted methods to initiate research, and another 7% use unequally weighted methods to interpret findings from the main study. The initiation design was used to help the research team to "set the foundation" and "gain a better understanding" of the context of the study before launching the main study. The interpretation design was implemented to "support some of our results" or confirm findings of the main study, sometimes at the request of editors and reviewers who asked for confirmatory evidence of findings.

Thirteen percent of the studies use multiple methods in a complementarity design. Methods were implemented concurrently, that is, "not in any particular sequence." Rather, an MMR design was chosen in anticipation of the need to provide a "strong test of the theory" or to explore a phenomenon from multiple viewpoints: "Multiple methods can lead to valuable insights as we found out that the views of suppliers and customers differ when defining solutions."

Most of the 37 authors who responded to the survey readily identified the weight and timing for each method. Only two authors were unsure about the weight assigned to each phase of the research, stating that "both were presented" and each had "different goals." While the majority of the studies were intentionally designed as multiple methods research, a few authors stated that the research design "evolved" into a multiple methods study, sometimes at the request of journal reviewers and editors.

Benefits and challenges

Two categories emerged from the thematic coding of benefits of conducting MMR (Table 1): (1) the fit between research questions and multiple methods and (2) the strength of findings. First, research questions often pointed to the need for MMR design: "The main benefit of using multiple methods is that it produces better research—because you can answer questions with one method that can't even be posed within the framework of another." Multiple methods were particularly helpful in exploratory research with a broad focus: "I view research as learning and with MMR I am really able to get a better perspective of the thing I am studying than if I do a single method." Johnson and Onwuegbuzie (2004) and Kirkwood and Campbell-Hunt (2007) both note that broader research questions can be answered with multiple methods as some findings that are masked by one method (e.g., in a survey)

can be uncovered with a different one (e.g., in depth interviews).

The second category was the strength of findings: "I prefer to use multiple methods if possible because it demonstrates the robustness of a finding and provides evidence of external validity." Triangulation of evidence was cited as a significant benefit of MMR: "The main attraction of the multiple-method approach is the opportunity for triangulation that it provides. Given the reassuring pattern of convergence we discovered, it enhanced the degree of confidence we have in our reported results." The robustness of findings is the most common benefit discussed in the literature on conducting MMR (Boyer and Swink 2009; Carter et al. 2008; Jick 1979; Johnson and Onwuegbuzie 2004; Kirkwood and Campbell-Hunt 2007). Due to the robustness of findings, multiple methods studies also tell a more convincing story: "[It] was a pretty compelling story to prove that the interplay between the qualitative and quantitative made a stronger argument for what we discovered and to show our colleagues and peers that these methods are compatible." Stewart (2009) also proposes that multiple methods findings are more compelling than single method outcomes.

It was not surprising to find that resources needed to conduct multiple methods studies represented a significant category of the challenges of MMR. The time commitment for MMR can be costly: "There was a significant increase in time to design and execute our study … our hunch is at least twice the norm." Finding the necessary expertise was also a resource issue: "Since research and researchers are becoming more specialized, MMR will likely require a team of researchers who are experts in different methodologies." Coordinating the work of a diverse team of researchers adds another layer of complexity to the research process that can lengthen the time devoted to a project. Similarly, Tashakkori and Teddlie (1998) cite the logistics of conducting multiple methods studies as a core concern in mixed methods research.

A second category reflected challenges in publishing multiple methods research. The apparent unfamiliarity with MMR on the part of some reviewers was cited as a significant obstacle: "Reviewers tend to focus on the method they like and suggest dropping studies that use methods they don't [like]. It is an uphill battle with reviewers." The review process was often frustrating because "reviewers have their own methods they feel comfortable with and see flaws in other methods as fatal. When using multiple methods, there is always something to see as fatal." The literature agrees that it is a challenge for scholars to have adequate understanding, let alone expertise, in a wide range of methodological approaches (Boyer and Swink 2009; Johnson and Onwuegbuzie 2004; Kirkwood and Campbell-Hunt 2007). Difficulties also arise when scholars in an area of research rely heavily on a single preferred method: "I found it impossible to get a field experiment ... published because editors sent it to experts on [the substantive topic] who like qualitative research." In addition, journal page constraints pose a challenge to MMR: "It is tough to conform to the page limitations while doing justice to both studies."

In addition to the challenges of conducting and publishing multiple methods studies, authors commented on the political aspects of implementing MMR. For example, engaging in MMR may adversely affect promotion and tenure decisions since "committees like to be able to position someone easily in one domain. People who use multiple methods suffer from a kind of 'conglomerate discount.' It is hard to have both breadth and impact." Multiple methods research maybe better "suited to more senior researchers who have already mastered one of the methods" and could be "risky for Ph.D. students and junior faculty" due to the increased time and higher skill levels required. Creswell (2009) extends political considerations to issues with securing funding for MMR projects.

Implications of findings

The aim of this research was to examine the implementation of multiple methods research in marketing. Findings show that multiple methods research is relatively rare in marketing, representing less than 4% of empirical studies published in top marketing journals over nearly two decades. While authors report significant benefits of MMR, they also describe challenges in conducting and publishing such research that may have a chilling effect its implementation in marketing. It is not our intent to advocate a widespread shift in the marketing discipline to multiple methods. However, as described in this paper, multiple methods research offers a promising response to critiques of method bias in marketing research. Our findings suggest several avenues for marketing researchers and scholars that could advance the implementation of MMR in the marketing discipline.

It would be instructive for authors of multiple methods studies to adopt a consistent approach in reporting their studies. First, authors should clearly identify the study as multiple methods, state the purpose of the MMR design and provide a clear description of the various methods and data sources employed. Second, 26 articles in the content analysis referred to preliminary studies that were neither fully described nor reported. It would have been helpful to have more information about those studies, including a description of the data sources, analysis methods, and, perhaps, a note that full results could be obtained from the authors. Finally, the advancement of MMR would benefit from the adoption of a common vocabulary in reporting multiple methods studies, which

would not only make the method clearer for the reader but also set a good example for other researchers who might consider implementing MMR designs. For example, it was sometimes difficult to classify the research purpose of a multiple methods study based on the description of the methodology provided. The weight and timing of methods were not described and were not evident in some studies. It could be that the weight shifted in response to reviewers' and editors' requests in the final revisions of manuscripts, as noted by some of the authors. However, it would be useful to report the original design in the methods section, even if the weight and timing are not maintained in the final revision.

Authors described resource constraints as a major challenge to conducting MMR. To address this concern, researchers could consider collaborating with coauthors to add the necessary methodological expertise. Given the prevalence of multiple authors for research published in the major marketing journals, it is surprising that we did not find more studies using MMR design. Team-based research is more likely to produce relevant, robust findings because it permits continuity beyond a single researcher, brings more resources to bear on research questions, allows gathering and analysis of more data, and affords multiple perspectives and skill sets (Pfeffer 2007).

Authors also identified concerns with reviewers in publishing MMR. The content analysis presented here identified a relatively small number of multiple methods studies published in the top marketing journals, and reviewers are often drawn from the pool of authors who publish in particular journals. Until there is a healthy representation of MMR in leading marketing journals, there will be a scarcity of reviewers who are capable of providing thoughtful reviews of multiple methods studies. While journal reviewers maybe competent in several methods, they may not be equipped to offer useful comments for improving MMR manuscripts, which bring into play an additional set of considerations beyond the rigor of individual methods. For example, do the authors justify the use of multiple methods design? Is the report of findings appropriate for the design? Editors could address this issue by developing a list of reviewers who are competent in evaluating MMR. At a minimum, journals could include multiple methods on the checklist of methodological expertise for potential reviewers.

As with any research design, this study has limitations that should be considered in interpreting the findings. Our sample was limited to five U.S.-based journals. We did not include international journals because these journals are not cited as frequently as those chosen (Tellis et al. 1999; Zinkhan 2003). In general, European journals are often more diverse in methods and may contain a different percentage of multiple methods studies, compared to those analyzed in this study. The impact of international journals is increasing, and they should be considered in future reviews (Stewart 2007).

The current study was limited to examining the application of multiple methods studies published in a single article; therefore, it offers no insight on the use of multiple methods in a research program. Thus, the use of MMR by a single researcher or research team across time is likely to be greater than is reflected in this study. For example, the initiation and development designs across studies in a research program maybe commonplace in marketing research. Further research could examine the extent to which marketing researchers employ multiple methods across more than one study in their research programs. In addition, an examination of the use of multiple methods in influential research streams, which was beyond the scope of this study, would be informative for the discipline. Meta-analyses that comment on the diversity of methods used to investigate key marketing phenomena, such as trust or market orientation, could point to areas with greater vulnerability to common method bias. Finally, the focus of this study was multiple methods research; therefore, we do not draw any conclusions about the relative strengths and weaknesses of this approach versus single method research design. Further research that addresses this issue would be beneficial to researchers as they weigh the costs and benefits of different research designs. Such research could include a more extensive collection and analysis of data from authors who employ both multiple methods and single method designs.

Multiple methods research offers a promising avenue for advancing the marketing discipline by providing robust findings that overcome the considerable risk of method bias. However, our enthusiasm is tempered by reports from authors of the challenges of conducting and publishing multiple methods studies. As noted by an author in response to the member check, "It is particularly difficult to go against the norm of an area, no matter what the method."

Table 1 Benefits and challenges of multiple methods research

Benefits
• Provides stronger results through triangulation of findings
• Can answer broader research questions
• Compensates for various weaknesses of single research methods
• Tells a more comprehensive, complete and convincing story
• Provides a holistic understanding of phenomena
Challenges
• Takes more resources (time, money, expertise) to conduct
• Requires understanding and training in multiple methods
• May require coordinating the work of a research team
• Can encounter difficulties in the review process
• Can have difficulty reporting results within journal page constraints

Appendix 1

Table 2 Multiple methods studies by research purpose

Year	Journal	Authors	Title
Development			
1990	*JCR* 16(4)	Bearden, William O. and Randall L. Rose	Attention to Social Comparison Information: An Individual Difference Factor Affecting Consumer Conformity
1991	*JCR* 18(1)	Richins, Marsha L.	Social Comparison and the Idealized Images of Advertising
1992	*JMR* 29(2)	Feinberg, Fred M., Barbara E. Kahn and Leigh McAlister	Market Share Response When Consumers Seek Variety
1992	*JCR* 19(1)	Rose, Randall L., William O. Bearden and Jesse E. Teel	An Attributional Analysis of Resistance to Group Pressure Regarding Illicit Drug and Alcohol Consumption
1992	*JCR* 19(1)	Marmorstein, Howard, Dhruv Grewal and Raymond P. H. Fishe	The Value of Time Spent in Price-Comparison Shopping: Survey and Experimental Evidence
1993	*JCR* 20(1)	Cole, Catherine A. and Siva K. Balasubramanian	Age Differences in Consumers' Search for Information: Public Policy Implications
1994	*JMR* 31(2)	Zenor, Michael J.	The Profit Benefits of Category Management
1994	*JMR* 31(2)	Dacin, Peter A. and Daniel C. Smith	The Effect of Brand Portfolio Characteristics on Consumer Evaluations of Brand Extensions
1994	*JCR* 20(4)	Tepper, Kelly	The Role of Labeling Processes in Elderly Consumers' Responses to Age Segmentation Cues
1994	*JCR* 20(4)	Tripp, Carolyn, Thomas D. Jensen and Les Carlson	The Effects of Multiple Product Endorsements by Celebrities on Consumers' Attitudes and Intentions
1994	*JCR* 21(1)	Park, C. Whan, David L. Mothersbaugh and Lawrence Feick	Consumer Knowledge Assessment
1994	*JCR* 21(3)	Richins, Marsha L.	Valuing Things: The Public and Private Meanings of Possessions
1994	*JCR* 21(3)	Richins, Marsha L.	Special Possessions and the Expression of Material Values
1995	*JCR* 22(3)	Rook, Dennis W. and Robert J. Fisher	Normative Influences on Impulsive Buying Behavior
1997	*MKS* 16(4)	West, Patricia M., Patrick L. Brockett and Linda L. Golden	A Comparative Analysis of Neural Networks and Statistical Methods for Predicting Consumer Choice
1998	*MKS* 17(4)	Wertenbroch, Klaus	Consumption Self-Control by Rationing Purchase Quantities of Virtue and Vice
1998	*JAMS* 26(3)	Hurley, Robert F. and Hooman Estelami	Alternative Indexes for Monitoring Customer Perceptions of Service Quality: A Comparative Evaluation in a Retail Context
1999	*JM* 63(2)	Menon, Anil, Sundar G. Bharadwaj, Phani Tej Adidam and Steven W. Edison	Antecedents and Consequences of Marketing Strategy Making: A Model and a Test
2000	*JMR* 37(2)	Dawar, Niraj and Madan M. Pillutla	Impact of Product-Harm Crises on Brand Equity: The Moderating Role of Consumer Expectations
2000	*JCR* 27(1)	Grayson, Kent and David Shulman	Indexicality and the Verification Function of Irreplaceable Possessions: A Semiotic Analysis
2001	*JCR* 28(1)	Inman, Jeffrey J.	The Role of Sensory-Specific Satiety in Attribute-Level Variety Seeking
2001	*JCR* 28(2)	Fitzsimmons, Gavan J. and Baba Shiv	Nonconscious and Contaminative Effects of Hypothetical Questions on Subsequent Decision Making
2001	*JM* 65(2)	Houston, Mark B., Beth A. Walker, Michael D. Hutt and Peter H. Reingen	Cross-Unit Competition for a Market Charter: The Enduring Influence of Structure
2001	*JM* 65(4)	Swaminathan, Vanitha, Richard J. Fox and Srinivas K. Reddy	The Impact of Brand Extension Introduction on Choice

Table 2 (continued)

Year	Journal	Authors	Title
2001	*JMR* 38(1)	Soman, Dilip and John T. Gourville	Transaction Decoupling: How Price Bundling Affects the Decision to Consume
2001	*MKS* 20(2)	Anderson, Eric T. and Duncan I. Simester	Are Sale Signs Less Effective When More Products Have Them?
2001	*MKS* 20(4)	Krider, Robert E., Priya Raghubir and Aradhna Krishna	Pizzas: _ or Square? Psychophysical Biases in Area Comparisons
2002	*JCR* 28(4)	Mothersbaugh, David L., Bruce A. Huhmann and George R. Franke	Combinatory and Separative Effects of Rhetorical Figures on Consumers' Effort and Focus in Ad Processing
2002	*JM* 66(1)	Lemon, Katherine N., Tiffany Barnett White and Russell S. Winer	Dynamic Customer Relationship Management: Incorporating Future Considerations into the Service Retention Decision
2002	*JM* 66(1)	McAlexander, James H., John W. Schouten and Harold F. Koenig	Building Brand Community
2002	*JM* 66(2)	Moore, Elizabeth S., William L. Wilkie and Richard J. Lutz	Passing the Torch: Intergenerational Influences as a Source of Brand Equity
2002	*JM* 66(3)	Balasubramanian, Siva K. and Catherine Cole	Consumers' Search and Use of Nutrition Information: The Challenge and Promise of the Nutrition Labeling and Education Act
2002	*JMR* 39(2)	Lemon, Katherine N. and Stephen M. Nowlis	Developing Synergies Between Promotions and Brands in Different Price-Quality Tiers
2002	*JMR* 39(4)	Mittal, Vikas, William T. Ross Jr. and Michael Tsiros	The Role of Issue Valence and Issue Capability in Determining Effort Investment
2002	*MKS* 21(1)	Soman, Dilip and Amar Cheema	The Effect of Credit on Spending Decisions: The Role of the Credit Limit and Credibility
2002	*MKS* 21(2)	Chen, Haipeng (Allan) and Akshay R. Rao	Close Encounters of Two Kinds: False Alarms and Dashed Hopes
2002	*MKS* 21(4)	Sayman, Serdar, Stephen J. Hoch and Jagmohan S. Raju	Positioning of Store Brands
2004	*JM* 68(4)	Lichtenstein, Donald R., Minette Drumwright and Bridgette M. Braig	The Effect of Corporate Social Responsibility on Customer Donations to Corporate-Supported Nonprofits
2005	*JAMS* 33(3)	Venable, Beverly T., Gregory M. Rose, Victoria D. Bush and Faye W. Gilbert	The Role of Brand Personality in Charitable Giving: An Assessment and Validation
2005	*JAMS* 33(4)	Johnson, Joseph and Gerard J. Tellis	Blowing Bubbles: Heuristics and Biases in the Run-Up of Stock Prices
2005	*JCR* 32(3)	Shrum, L.J., James E. Burroughs and Aric Rindfleisch	Television's Cultivation of Material Values
2005	*JMR* 42(1)	Okada, Erica Mina	Justification Effects on Consumer Choice of Hedonic and Utilitarian Goods
2005	*JMR* 42(4)	Jedidi, Kamel and Rajeev Kohli	Probabilistic Subset-Conjunctive Models for Heterogeneous Consumers
2005	*MKS* 24(2)	Leclerc, France, Christopher K. Hsee, and Joseph C. Nunes	Narrow Focusing: Why the Relative Position of a Good in Its Category Matters More Than It Should
2005	*MKS* 24(3)	Gourville, John T. and Dilip Soman	Overchoice and Assortment Type: When and Why Variety Backfires
2006	*JCR* 33(3)	Abendroth, Lisa J. and Kristin Diehl	Now or Never: Effects of Limited Purchase Opportunities on Patterns of Regret over Time
2006	*JCR* 33(3)	Sood, Sanjay and Xavier Dreze	Brand Extensions of Experiential Goods: Movie Sequel Evaluations
2006	*JM* 70(1)	Antia, Kersi D., Mark E. Bergen, Shantanu Dutta and Robert J. Fisher	How Does Enforcement Deter Gray Market Incidence?
2007	*JM* 71(1)	Reed II, Americus, Karl Aquino and Eric Levy	Moral Identity and Judgments of Charitable Behavior
2007	*JMR* 44(2)	Herzenstein, Michal, Steven S. Posavac and J. Joško Brakus	Adoption of New and Really New Products: The Effects of Self-Regulation Systems and Risk Salience
2007	*JMR* 44(3)	Dahl, Darren W. and C. Page Moreau	Thinking Inside the Box: Why Consumers Enjoy Constrained Creative Experiences
2007	*MKS* 26(4)	Berger, Jonah, Michaela Draganska and Itamar Simonson	The Influence of Product Variety on Brand Perception and Choice
2008	*JMR* 45(1)	Berger, Jonah and Gráinne Fitzsimons	Dogs on the Street, Pumas on Your Feet: How Cues in the Environment Influence Product Evaluation and Choice

Table 2 (continued)

Year	Journal	Authors	Title
2008	*JMR* 45(2)	Hsee, Christopher K., Jean-Pierre Dubé and Yan Zhang	The Prominence Effect in Shanghai Apartment Prices
2008	*JMR* 45(2)	Lee, Hyun-Jung, Jongwon Park, Jin-Yong Lee and Robert S. Wyer Jr	Disposition Effects and Underlying Mechanisms in E-Trading of Stocks
2008	*JMR* 45(3)	He, Xin, J. Jeffrey Inman and Vikas Mittal	Gender Jeopardy in Financial Risk Taking
2008	*JAMS* 36(4)	Jones, Tim, Shirley F. Taylor and Harvir S. Bansal	Commitment to a Friend, a Service Provider, or a Service Company—Are They Distinctions Worth Making?
2008	*JCR* 35(3)	Du, Shuili, Sankar Sen and C.B. Bhattacharya	Exploring the Social and Business Returns of a Corporate Oral Health Initiative Aimed at Disadvantaged Hispanic Families
Initiation			
1991	*MKS* 10(4)	Kahn, Barbara E. and Jagmohan S. Raju	Effects Of Price Promotions On Variety-Seeking And Reinforcement Behavior
1992	*JMR* 29(1)	Buchanan, Lauranne	Vertical Trade Relationships: The Role of Dependence and Symmetry in Attaining Organizational Goals
1993	*JM* 57(3)	Loken, Barbara and Deborah Roedder John	Diluting Brand Beliefs: When Do Brand Extensions Have a Negative Impact?
1994	*JCR* 20(4)	Babin, Barry J., William R. Darden and Mitch Griffin	Work and/or Fun: Measuring Hedonic and Utilitarian Shopping Value
1997	*JM* 61(2)	Song, X. Michael and Mark E. Parry	A Cross-National Comparative Study of New Product Development Processes: Japan and the United States
1997	*JMR* 34(1)	Song, X. Michael and Mark E. Parry	The Determinants of Japanese New Product Successes
1998	*JAMS* 26(2)	Gwinner, Kevin P., Dwayne D. Gremler and Mary Jo Bitner	Relational Benefits in Services Industries: The Customer's Perspective
1998	*JAMS* 26(2)	Hurley, Robert F.	Customer Service Behavior in Retail Settings: A Study of the Effect of Service Provider Personality
2000	*JMR* 37(4)	Homburg, Christian and Christian Pflesser	A Multiple-Layer Model of Market-Oriented Organizational Culture: Measurement Issues and Performance Outcomes
2002	*JMR* 39(1)	Dahl, Darren W. and Page Moreau	The Influence and Value of Analogical Thinking During New Product Ideation
2002	*JMR* 39(3)	Chandon, Pierre and Brian Wansink	When Are Stockpiled Products Consumed Faster? A Convenience-Salience Framework of Postpurchase Consumption Incidence and Quantity
2002	*MKS* 21(3)	Van Herpen, Erica and Rik Pieters	The Variety of an Assortment: An Extension to the Attribute-Based Approach
2003	*JM* 67(3)	Sprott, David E., Kenneth C. Manning and Anthony D. Miyazaki	Grocery Price Setting and Quantity Surcharges
2004	*JM* 68(3)	Dellande, Stephanie, Mary C. Gilly and John L. Graham	Gaining Compliance and Losing Weight: The Role of the Service Provider in Health Care Services
2005	*MKS* 24(1)	Montgomery, David B., Marian Chapman Moore and Joel E. Urbany	Reasoning About Competitive Reactions: Evidence from Executives
2006	*JAMS* 34(2)	Ellen, Pam Scholder, Deborah J. Webb and Lois A. Mohr	Building Corporate Associations: Consumer Attributions for Corporate Socially Responsible Programs
2006	*JM* 70(1)	Ulaga, Wolfgang and Andreas Eggert	Value-Based Differentiation in Business Relationships: Gaining and Sustaining Key Supplier Status
2007	*JCR* 34(3)	Chandon, Pierre and Brian Wansink	The Biasing Health Halos of Fast-Food Restaurant Health Claims: Lower Calorie Estimates and Higher Side-Dish Consumption Intentions
2007	*JM* 71(1)	Quellet, Jean-François	Consumer Racism and Its Effects on Domestic Cross-Ethnic Product Purchase: An Empirical Test in the United States, Canada, and France

Table 2 (continued)

Year	Journal	Authors	Title
Complementarity			
1990	*JAMS* 18(1)	Ram, S. and Hyung-Shik Jung	The Conceptualization and Measurement of Product Usage
1992	*JCR* 19(1)	Burke, Raymond R., Bari A. Harlam, Barbara E. Kahn and Leonard M. Lodish	Comparing Dynamic Consumer Choice in Real and Computer-simulated Environments
1993	*JCR* 20(1)	Arnould, Eric J. and Linda L. Price	River Magic: Extraordinary Experience and the Extended Service Encounter
1995	*JAMS* 23(3)	Kitchell, Susan	Corporate Culture, Environmental Adaptation, and Innovation Adoption: A Qualitative/Quantitative Approach
1996	*MKS* 15(2)	Lenk, Peter J., Wayne S. Desarbo, Paul E. Green and Martin R. Young	Hierarchical Bayes Conjoint Analysis: Recovery of Partworth Heterogeneity from Reduced Experimental Designs
1996	*JCR* 22(4)	Sirsi, Ajay K., James C. Ward and Peter H. Reingen	Microcultural Analysis of Variation in Sharing of Causal Reasoning about Behavior
1999	*JM* 63(3)	Clark, Bruce H. and David B. Montgomery	Managerial Identification of Competitors
1999	*JM* 63(4)	Price, Linda L. and Eric J. Arnould	Commercial Friendships: Service Provider-Client Relationships in Context
2002	*JMR* 39(4)	MacInnis, Deborah J., Ambar G. Rao and Allen M. Weiss	Assessing When Increased Media Weight of Real-World Advertisements Helps Sales
2003	*JCR* 30(2)	Coulter, Robin A., Linda L. Price and Lawrence Feick	Rethinking the Origins of Involvement and Brand Commitment: Insights from Postsocialist Central Europe
2007	*JCR* 34(2)	Thompson, Craig J. and Gokcen Coskuner-Balli	Countervailing Market Responses to Corporate Co-optation and the Ideological Recruitment of Consumption Communities
2007	*JM* 71(3)	Tuli, Kapil R., Ajay K. Kohli and Sundar G. Bharadwaj	Rethinking Customer Solutions: From Product Bundles to Relational Processes
Interpretation			
2002	*JAMS* 30(4)	Reibstein, David J.	What Attracts Customers to Online Stores, and What Keeps Them Coming Back?
2002	*JCR* 29(2)	Morrin, Maureen, Jacob Jacoby, Gita Venkataramani Johar, Xin He, Alfred Kuss and David Mazursky	Taking Stock of Stockbrokers: Exploring Momentum versus Contrarian Investor Strategies and Profiles
2002	*JMR*39(2)	Heilman, Carrie M., Kent Nakamoto and Ambar G. Rao	Pleasant Surprises: Consumer Response to Unexpected In-Store Coupons
2004	*JAMS* 32(3)	Jayachandran, Satish, Kelly Hewett and Peter Kaufman	Customer Response Capability in a Sense-and-Response Era: The Role of Customer Knowledge Process
2005	*JCR* 32(2)	Commuri, Suraj and James W. Gentry	Resource Allocation in Households with Women as Chief Wage Earners
2007	*JCR* 34(3)	Scott, Linda and Patrick Vargas	Writing with Pictures: Toward a Unifying Theory of Consumer Response to Images

Appendix 2

Table 3 Coding scheme

Theme/Node	Meaning	Text Coded
Methods	Methods used for the various studies	Quantitative and qualitative methods stated by respondent
Weight	The relative priority assigned to the methods	Equal or unequal weight assigned by respondent
Timing	The order of implementation of the methods and/or use of the data in analysis	Sequential or concurrent use of data reported by respondent
Data Collection Benefits	Benefits of MMR related to data collection	Positive aspects of collecting data
Data Analysis Benefits	Benefits of MMR related to data analysis	Positive aspects of analyzing data
Publishing Benefits	Benefits of MMR related to publishing	Positive aspects of publishing the study
Data Collection Challenges	Issues with data collection	Negative aspects of collecting data
Data Analysis Challenges	Issues with data analysis	Negative aspects of analyzing data
Publishing Challenges	Issues with publishing	Negative aspects of publishing the study
Observations	Additional observations about conducting MMR	Additional information provided regarding conducting MMR

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